Bloch's Theorem Explained Simply

An illustrated reading companion

Based on the video by Phanimations (YouTube, 13 July 2024)

https://www.youtube.com/watch?v=85bdcSgdXIc

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These notes follow the order of the video. The text restates what the lecturer says, in prose. Passages marked ***Correction note*** *are additions by the note-taker. They clarify places where the video's argument is loose or incomplete, and they are not part of the video.*

Introduction

Bloch's theorem is a very powerful tool in quantum and condensed matter physics. It tells us how electrons, and other wave-like particles, behave in a periodic crystal lattice. The topic is a little advanced, but a basic knowledge of quantum mechanics and an understanding of Euler's formula should be enough to follow the argument.

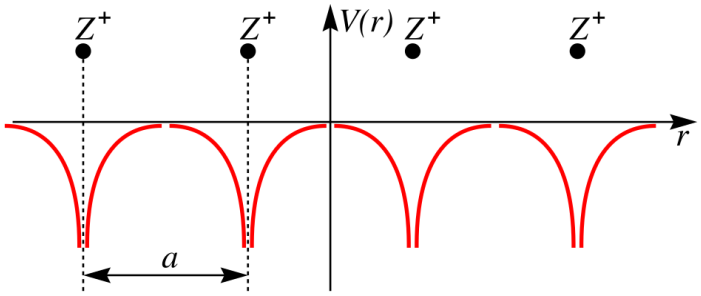


Figure 1: Qualitative potential energy $V(r)$ of an electron along a chain of positive ions (Z^+) spaced by the lattice constant a. The potential repeats from cell to cell, which is the periodic environment Bloch's theorem is about. *Image:* Potential-actual.svg, CC0.

The Crystal Lattice and Translational Symmetry

An infinite line of points

Imagine a line of dots that stretches out to infinity both to the left and to the right. Because the line is infinite, moving from one dot to another makes literally no difference. You could not tell which dot you are at, or whether you are in the middle or at an end, because there is no end.

Such a system does not exist in real life, but it can be approximated by a very long line of dots. As long as you are somewhere in the middle, moving from one dot to the next makes an almost negligible difference.

Two and three dimensions

The same idea scales up to two dimensions: an infinite plane of dots, or at least a very large one. There is a definite distance between the dots, and that distance can be changed without changing the overall shape of the structure or its periodicity. Such a structure is called a *lattice*, or *crystal lattice*.

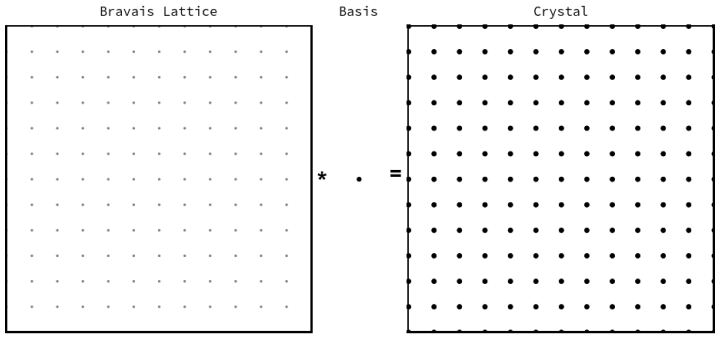


Figure 2: Combining a two-dimensional lattice of points with a basis gives a crystal. The lattice spacing can change without changing the periodic pattern. *Image:* Crystal Structure as Lattice and Basis.gif, CC0.

Adding one more dimension gives a three-dimensional lattice, which is what most real crystals are like. The 3D crystal has the same properties as before:

- a constant distance between points, from now on called the **lattice constant** a;
- **translational symmetry**. Moving from one point to another is the same as moving by an integer multiple of the lattice constant, and after such a move the crystal looks unchanged from the inside.

Any discrepancies visible in a model crystal arise simply because it is not infinite. This is not much of an issue in practice, because most crystals contain trillions upon trillions of lattice sites (also called unit cells), and most forces dissipate quite quickly with distance.

> **Correction note:** The Coulomb force between charges is not short-ranged; it falls off only as $1/r^2$. Surface effects are negligible for bulk properties mainly because only a tiny fraction of the atoms lies at the surface (roughly $N^{-1/3}$ of N atoms). Screening by the other electrons also helps.

Back to one dimension

Everything that follows can be extended to two or three dimensions. One dimension, however, is much easier both to visualise and to handle mathematically. The lattice constant a can be scaled up or down, but it is a constant: the same distance separates all neighbouring lattice points.

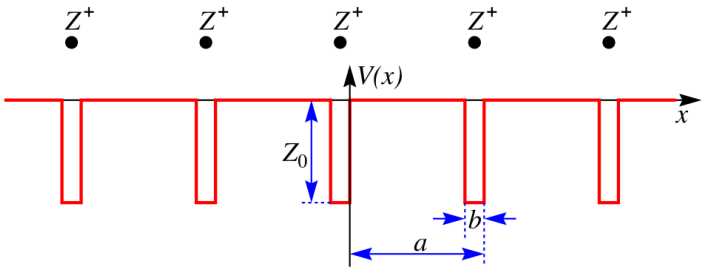


Figure 3: A one-dimensional lattice idealised as a periodic square-well potential. Ions Z^+ repeat every lattice constant a, so a shift by a leaves the lattice unchanged. *Image:* Periodic square potential 130707.svg, CC0.

Certain functions behave in a similar way. A sine wave, for example, repeats itself after a fixed shift. The formal term for this property is *translational symmetry.*

The Quantum-Mechanical Foundation

Why the wave function must reflect the lattice

Crystals are typically made of atoms or molecules, which are small enough that quantum mechanics is very relevant. Quantum mechanics brings with it the all-important wave function $\psi(x)$. Since the crystal lattice is periodic and has translational symmetry, the wave function should also be periodic. More precisely, the squared magnitude of the wave function should be periodic:

$$|\psi(x+a)|^2 = |\psi(x)|^2.$$

The video presents this simply as a law of quantum mechanics.

Correction note: A periodic $|\psi|^2$ on its own does not force $\psi(x+a) = C\,\psi(x)$. Nor is it a general law. A superposition of two states with different k, for example, has a non-periodic $|\psi|^2$. The rigorous starting point is that the Hamiltonian $H = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x)$ has a periodic potential, $V(x+a) = V(x)$, so H commutes with the translation operator $T_a\psi(x) = \psi(x+a)$. Energy eigenstates can therefore be chosen to be eigenstates of T_a as well, which gives $\psi(x+a) = C\,\psi(x)$. Because T_a is unitary (and normalisable under the periodic boundary condition used below), $|C| = 1$. A periodic $|\psi|^2$ is then a consequence for these states, not an assumption.

Allowing a phase factor

Because the quantum wave function is complex-valued, we can place a coefficient C on one side and drop the magnitudes:

$$\psi(x+a) = C\,\psi(x).$$

The condition is that the magnitude of this constant must be one, $|C| = 1$. Anyone who knows some complex analysis will recognise that C must then have the form

$$C = e^{i\theta}.$$

This equation will be very important. It gives us some generality, but it leaves a question: what is θ?

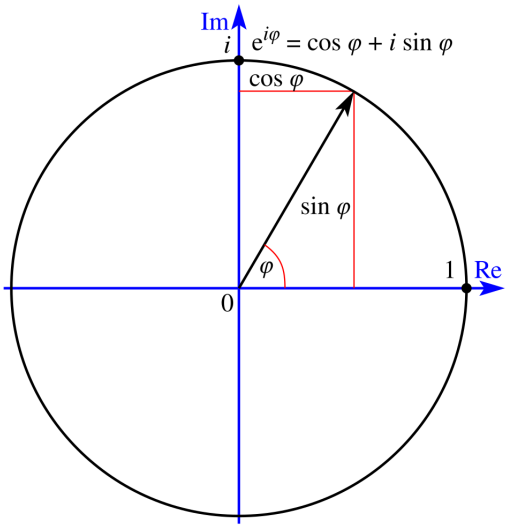


Figure 4: Euler's formula on the complex unit circle. Any number of unit magnitude is $e^{i\varphi} = \cos\varphi + i\sin\varphi$, which is why the constant must be $C = e^{i\theta}$. *Image:* Euler's formula.svg, CC BY-SA 3.0.

The Circular Crystal Model

A deliberately "ridiculous" assumption

To answer this question, we make a seemingly ridiculous assumption: the crystal is shaped like a circle. This circular crystal keeps the property of translational symmetry. It also has what are called

periodic boundary conditions: if you travel once around the circle, you end up exactly where you started. This is the property we will exploit.

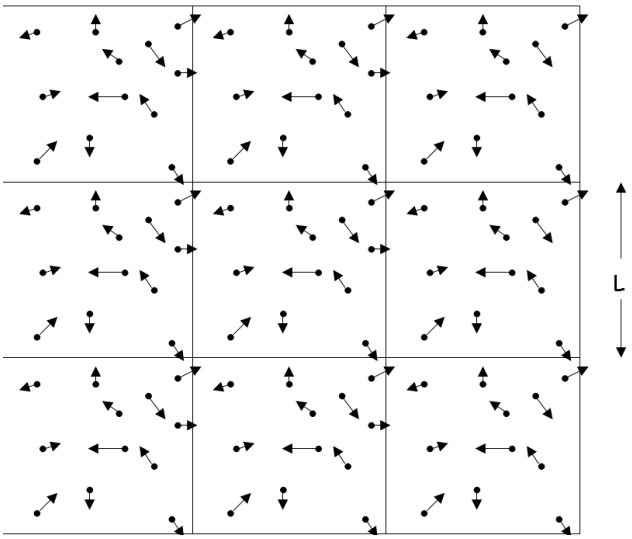


Figure 5: Periodic boundary conditions, known in textbooks as Born–von Kármán conditions. A finite block of crystal is treated as if it were repeated endlessly, so leaving through one edge brings you back through the opposite edge. This is the same idea as the video's circular crystal. *Image:* Born Von Karman.png, Public domain.

Why the assumption is justified

The justification comes down to how many atoms most crystals contain. As the radius of the circle and the number of points on it both increase, the edge of the circle becomes closer and closer to a straight line, and in the limit of infinite radius it actually becomes one. Such a system does not exist in real life either, but with as many atoms as most crystals contain, a crystal can be approximated as a very big circle. In more than one dimension the shape would actually be a torus.

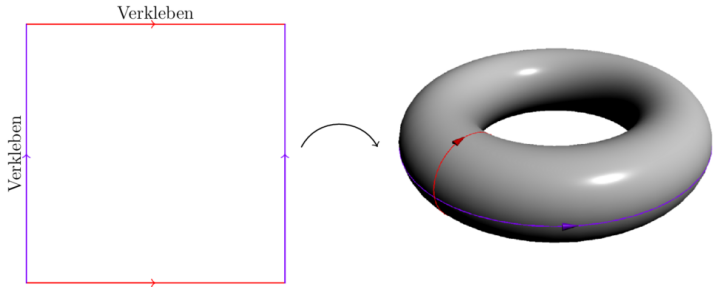


Figure 6: Gluing the opposite edges of a square gives a torus, the two-dimensional version of the ring-shaped crystal mentioned in the video. The German label "Verkleben" on both edges means "glue". Edges with matching colours and arrows are glued together: red to red, which gives the red loop, and violet to violet, which gives the violet loop. *Image:* Ebene zu Torus verkleben 01.png, CC BY-SA 4.0.

The boundary condition

Written in terms of the wave function, the periodic boundary condition reads

$$\psi(x) = \psi(x + Na),$$

where Na means moving by the number of lattice sites, N, times the lattice constant. This is clearly the same as going all the way around the circle. By the same logic, N applications of the factor C must give back the original wave function:

$$C^N = 1.$$

Correction note: In textbooks this ring trick is called the **Born–von Kármán** periodic boundary condition. It is a mathematical convenience; bulk results do not depend on it when N is large.

Placing the ring on the complex unit circle

Now place the circular crystal on top of the unit circle in the complex plane. Moving from one lattice site to the next is then the same as rotating by some angle θ in the complex plane, and that rotation is the same as multiplying by $e^{i\theta}$.

Because all the angular spacings are equal, each angle can be written as 2π times some integer, divided by the number of lattice sites N:

$$\theta = \frac{2\pi s}{N}, \qquad s \in \mathbb{Z}.$$

The numbers $e^{i\theta}$ are therefore what are called the **N-th roots of unity**:

$$C = e^{i2\pi s/N}.$$

This is the solution. The N-th roots of unity meet both requirements for C: they have magnitude one, and raising any of them to the power N always gives one.

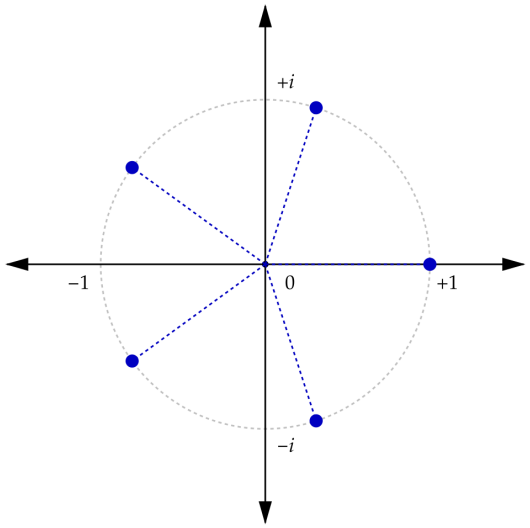


Figure 7: The fifth roots of unity ($N = 5$) are equally spaced points $e^{i2\pi s/N}$ on the unit circle. For a ring of N lattice sites, the Bloch phase factor C must be one of these N points. *Image:* One5Root.svg, CC BY-SA 3.0.

Correction note: The ring-on-unit-circle overlay is a mnemonic, not a proof. The lattice sites live in real space and the phase angles live in the complex plane, which are different spaces. The actual derivation is the Born–von Kármán condition $C^N = 1$. The result follows directly from algebra. $C^N = 1$ with $C = e^{i\theta}$ requires $N\theta = 2\pi s$, so $\theta = 2\pi s/N$. Laying the real-space ring over the complex unit circle is only a visual aid. For $s = 1$, one lattice step corresponds to one site-spacing around the circle. For general s, it corresponds to s site-spacings.

Deriving Bloch's Theorem

Returning to the original equations, we now know θ. The constant is $C = e^{i2\pi s/N}$, where s ranges from 0 to $N - 1$, because $s = N$ would be the same as $s = 0$. Hence

$$\psi(x + a) = e^{i2\pi s/N}\,\psi(x).$$

This is Bloch's theorem. It is usually written as

$$\psi(x + a) = e^{ika}\,\psi(x), \qquad k = \frac{2\pi s}{Na}.$$

The reason for writing it this way is touched on later in the video, and the full reason is left for another video. Note also that this is only the one-dimensional version.

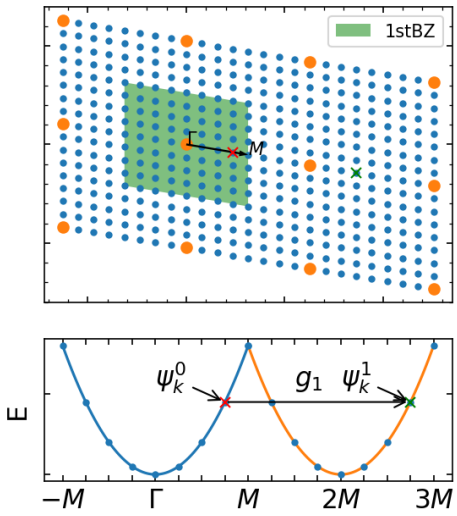


Figure 8: Wave vectors that differ by a reciprocal-lattice vector (g_1 in the figure) give equivalent Bloch states. This matches the remark that $s = N$ gives the same phase factor as $s = 0$, and it previews the reciprocal-space ideas promised for the sequel. *Image:* Brouillian zone, Bloch states.png, CC BY-SA 4.0.

Correction note: In the auto-captions the shifted argument is garbled ("x + sa a"). The single-step statement is $\psi(x + a) = e^{ika}\psi(x)$. Applying it n times gives $\psi(x + na) = e^{ikna}\psi(x)$. Also, the range $s = 0, \ldots, N - 1$ puts k in $[0, 2\pi/a)$. The usual convention is the first Brillouin zone $-\pi/a < k \leq \pi/a$. The two choices are equivalent, because k and $k + 2\pi/a$ give the same phase factor.

The second form of the theorem

A different representation of Bloch's theorem can be derived from these results. It tells us exactly what form $\psi(x)$ takes. A natural guess would be e^{ikx}, and this does meet the boundary conditions. The true answer is close: it is e^{ikx} times a new function $u(x)$,

$$\psi(x) = e^{ikx}\, u(x), \qquad u(x+a) = u(x),$$

where $u(x)$ has the periodicity of the lattice. It can be thought of as a **unit-cell function**. The sinusoid-like curve drawn in the video is representative of such a function.

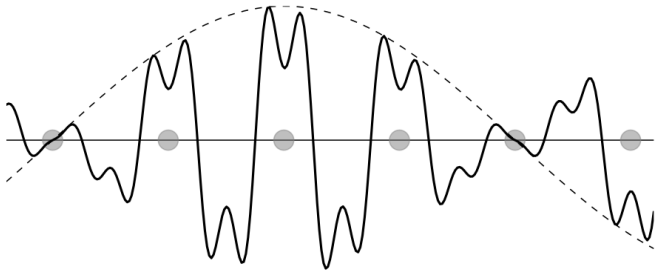


Figure 9: A one-dimensional Bloch wave. The real part (solid line) is the product of the plane-wave factor e^{ikx} (real part shown dotted, as the envelope) and a lattice-periodic function $u(x)$. Grey circles mark the atoms. *Image:* Bloch function.svg, CC BY 4.0.

Correction note: The video states this form without the derivation. It takes one line. Define $u(x) = e^{-ikx}\psi(x)$; then $u(x+a) = e^{-ik(x+a)}e^{ika}\psi(x) = u(x)$. The plane wave e^{ikx} is not wrong. It is the special case $u = \text{const}$, which is the free electron with $V = 0$.

Visualising Bloch Waves

The helix picture

Picture three axes: Re and Im, the real and imaginary number lines that form the complex plane, and x, the line along which the crystal lattice lies. Along x we draw the unit-cell function $u(x)$ together with e^{ikx}.

Think of e^{ikx} as an *envelope* to which $u(x)$ conforms. The result looks like a helix: the unit-cell function has been wrapped into the plane of e^{ikx}.

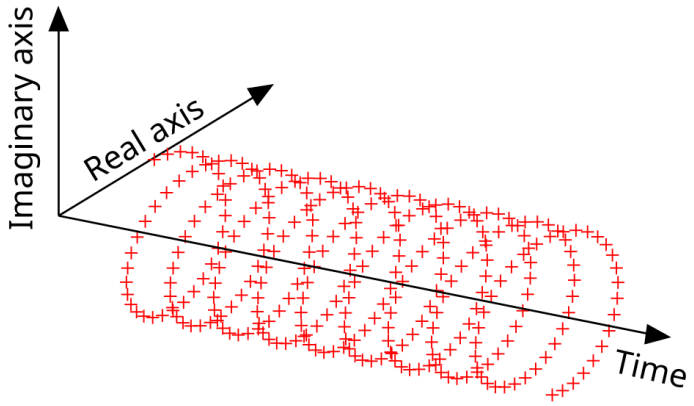


Figure 10: A complex exponential plotted along a third axis traces a helix in the (Re, Im) plane. Here the third axis is time; in the video it is the lattice coordinate x, and e^{ikx} is the envelope onto which $u(x)$ is wrapped. *Image:* Complex sinusoid 3D.svg, Public domain.

In this picture $u(x)$ is drawn as real-valued, but that need not be the case. If $u(x)$ is complex-valued, it is not wrapped so neatly. Instead, every point is rotated by the angle kx that corresponds to its position.

A phase that rotates, a magnitude that stays

Now take a single point (a blue dot in the animation) and move it along x by an integer multiple of the lattice constant a. Each time, the vector from the dot to the x-axis changes: it rotates. Its magnitude, however, remains the same.

Summary of the Theorem

The full Bloch theorem says that the electron wave function can be written as a complex exponential times a periodic function $u(x)$:

$$\psi(x) = e^{ikx}\, u(x), \qquad u(x+a) = u(x).$$

Because of this, whenever the wave function is translated by an integer multiple of the lattice constant, it differs only by a phase factor, and that factor happens to be an N-th root of unity.

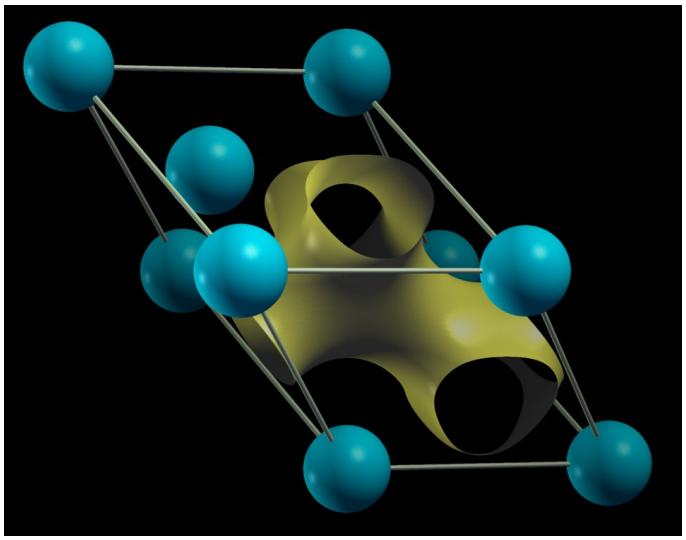


Figure 11: Bloch's theorem in three dimensions: an isosurface of $|\psi|^2$ for a Bloch state of silicon (lowest valence band at the Γ point). The density repeats from unit cell to unit cell. *Image:* BlochWave in Silicon.png, CC BY-SA 3.0.

Key Takeaways and Next Steps

- The electron wave function depends on a periodic "unit-cell function" $u(x)$.
- Whenever you shift by the lattice constant, the wave function picks up a phase factor: a shift by a multiplies ψ by e^{ika}, and a shift by an integer multiple na multiplies it by e^{ikna}. (The video says only "e^{ika}" for both cases.)
- The number k bears a striking resemblance to the k of a sinusoidal plane wave, $k = 2\pi/\lambda$. The two are indeed closely related. How this k relates to the **crystal momentum** and to **reciprocal space** is left for the sequel to this video.

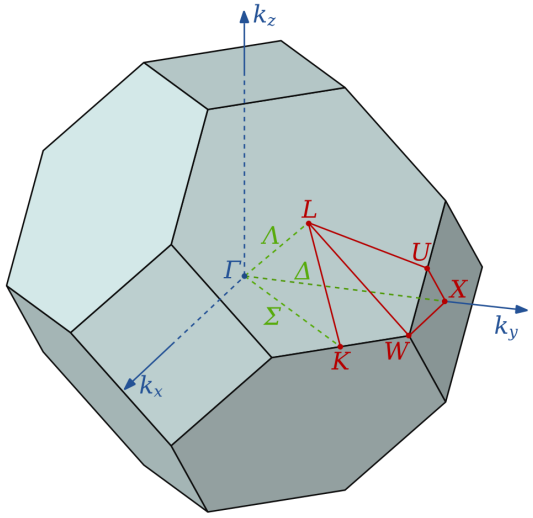


Figure 12: Outlook on reciprocal space: the first Brillouin zone of a face-centred cubic lattice, with high-symmetry points Γ, X, L, K, W and U marked. The crystal momentum k in Bloch's theorem lives in this zone. *Image:* Brillouin Zone (1st, FCC).svg, Public domain.

The creator notes that this is his first fairly big video made with Manim. He also says that he glossed over several things, because the main purpose was to build intuition for what Bloch's theorem means.

Rigorous derivations are available elsewhere, and he links a few in the video description: lectures by Jordan Edmunds and David Miller, notes from Binghamton University (https://bingweb.binghamton.edu/~suzuki/QuantumMechanicsFiles/12-5_Bloch_theorem.pdf), and the Manim source code for the animations (https://github.com/shawnseanspencer/manimations).

Image Credits and Sources

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